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== Vitis HLS Report for 'matVecMul'  
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\* Date: Tue May 5 15:09:25 2026

\* Version: 2021.1 (Build 3247384 on Thu Jun 10 19:36:07 MDT 2021)

\* Project: matVecMul\_stream\_prj

\* Solution: solution\_multivec\_bothpipelined (Vivado IP Flow Target)

\* Product family: zynq

\* Target device: xc7z020-clg400-1

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== Performance Estimates  
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+ Timing:

\* Summary:

| Clock  | Target   | Estimated | Uncertainty |
|--------|----------|-----------|-------------|
| ap_clk | 10.00 ns | 5.831 ns  | 2.70 ns     |

+ Latency:

\* Summary:

| Latency (cycles) |     | Latency (absolute) |     | Interval |     | Pipeline |
|------------------|-----|--------------------|-----|----------|-----|----------|
| min              | max | min                | max | min      | max | Type     |
| ?                | ?   | ?                  | ?   | ?        | ?   | no       |

+ Detail:

\* Instance:

| Instance                                        | Module                               | Latency (cycles) |     | Latency (absolute) |          | Interval |     | Pipeline |
|-------------------------------------------------|--------------------------------------|------------------|-----|--------------------|----------|----------|-----|----------|
|                                                 |                                      | min              | max | min                | max      | min      | max | Type     |
| grp_matVecMul_Pipeline_VITIS_LOOP_28_1_fu_219   | matVecMul_Pipeline_VITIS_LOOP_28_1   | 66               | 66  | 0.660 us           | 0.660 us | 66       | 66  | no       |
| grp_matVecMul_Pipeline_matmul_inner_loop_fu_240 | matVecMul_Pipeline_matmul_inner_loop | 70               | 70  | 0.700 us           | 0.700 us | 70       | 70  | no       |

\* Loop:

| Loop Name           | Latency (cycles) |      | Iteration | Initiation | Interval | Trip  |           |
|---------------------|------------------|------|-----------|------------|----------|-------|-----------|
|                     | min              | max  | Latency   | achieved   | target   | Count | Pipelined |
| - sample_loop       | ?                | ?    | 4871      | -          | -        | ?     | no        |
| + matmul_outer_loop | 4800             | 4800 | 75        | -          | -        | 64    | no        |

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== Utilization Estimates  
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\* Summary:

| Name       | BRAM_18K | DSP | FF  | LUT | URAM |
|------------|----------|-----|-----|-----|------|
| DSP        | -        | -   | -   | -   | -    |
| Expression | -        | -   | 0   | 294 | -    |
| FIFO       | -        | -   | -   | -   | -    |
| Instance   | 4        | 1   | 392 | 673 | -    |
| Memory     | 1        | -   | 0   | 0   | 0    |

|                 |     |     |        |       |   |
|-----------------|-----|-----|--------|-------|---|
| Multiplexer     | -   | -   | -      | 132   | - |
| Register        | -   | -   | 279    | -     | - |
| Total           | 5   | 1   | 671    | 1099  | 0 |
| Available       | 280 | 220 | 106400 | 53200 | 0 |
| Utilization (%) | 1   | ~0  | ~0     | 2     | 0 |

+ Detail:

\* Instance:

| Instance                                        | Module                               | BRAM_18K | DSP | FF  | LUT | URAM |
|-------------------------------------------------|--------------------------------------|----------|-----|-----|-----|------|
| control_s_axi_U                                 | control_s_axi                        | 4        | 0   | 230 | 302 | 0    |
| grp_matVecMul_Pipeline_VITIS_LOOP_28_1_fu_219   | matVecMul_Pipeline_VITIS_LOOP_28_1   | 0        | 0   | 9   | 72  | 0    |
| grp_matVecMul_Pipeline_matmul_inner_loop_fu_240 | matVecMul_Pipeline_matmul_inner_loop | 0        | 1   | 153 | 299 | 0    |
| Total                                           |                                      | 4        | 1   | 392 | 673 | 0    |

\* DSP:

N/A

\* Memory:

| Memory          | Module        | BRAM_18K | FF | LUT | URAM | Words | Bits | Banks | W*Bits*Banks |
|-----------------|---------------|----------|----|-----|------|-------|------|-------|--------------|
| x_localbuff_V_U | x_localbuff_V | 1        | 0  | 0   | 0    | 64    | 16   | 1     | 1024         |
| Total           |               | 1        | 0  | 0   | 0    | 64    | 16   | 1     | 1024         |

\* FIFO:

N/A

\* Expression:

| Variable Name              | Operation | DSP | FF | LUT | Bitwidth P0 | Bitwidth P1 |
|----------------------------|-----------|-----|----|-----|-------------|-------------|
| add_ln27_fu_274_p2         | +         | 0   | 0  | 38  | 31          | 1           |
| add_ln37_fu_285_p2         | +         | 0   | 0  | 39  | 32          | 1           |
| p_Val2_s_fu_353_p2         | +         | 0   | 0  | 23  | 16          | 16          |
| sub_fu_251_p2              | +         | 0   | 0  | 39  | 32          | 2           |
| and_ln789_fu_439_p2        | and       | 0   | 0  | 2   | 1           | 1           |
| and_ln790_fu_453_p2        | and       | 0   | 0  | 2   | 1           | 1           |
| and_ln795_fu_483_p2        | and       | 0   | 0  | 2   | 1           | 1           |
| carry_1_fu_373_p2          | and       | 0   | 0  | 2   | 1           | 1           |
| overflow_fu_477_p2         | and       | 0   | 0  | 2   | 1           | 1           |
| underflow_fu_501_p2        | and       | 0   | 0  | 2   | 1           | 1           |
| Range1_all_ones_fu_405_p2  | icmp      | 0   | 0  | 12  | 14          | 2           |
| Range1_all_zeros_fu_411_p2 | icmp      | 0   | 0  | 12  | 14          | 1           |
| Range2_all_ones_fu_389_p2  | icmp      | 0   | 0  | 12  | 13          | 2           |
| icmp_ln27_fu_269_p2        | icmp      | 0   | 0  | 18  | 32          | 32          |
| icmp_ln37_fu_280_p2        | icmp      | 0   | 0  | 18  | 32          | 32          |
| out_pkt_last_V_fu_295_p2   | icmp      | 0   | 0  | 18  | 32          | 32          |
| or_ln384_fu_515_p2         | or        | 0   | 0  | 2   | 1           | 1           |
| or_ln794_fu_465_p2         | or        | 0   | 0  | 2   | 1           | 1           |
| or_ln795_fu_489_p2         | or        | 0   | 0  | 2   | 1           | 1           |
| deleted_ones_fu_445_p3     | select    | 0   | 0  | 2   | 1           | 1           |
| deleted_zeros_fu_417_p3    | select    | 0   | 0  | 2   | 1           | 1           |
| select_ln384_fu_507_p3     | select    | 0   | 0  | 17  | 1           | 15          |
| val_V_fu_521_p3            | select    | 0   | 0  | 16  | 1           | 16          |
| xor_ln416_fu_367_p2        | xor       | 0   | 0  | 2   | 1           | 2           |
| xor_ln789_fu_433_p2        | xor       | 0   | 0  | 2   | 1           | 2           |
| xor_ln794_1_fu_471_p2      | xor       | 0   | 0  | 2   | 1           | 2           |

|                     |     |   |   |     |     |     |
|---------------------|-----|---|---|-----|-----|-----|
| xor_ln794_fu_459_p2 | xor | 0 | 0 | 2   | 1   | 2   |
| xor_ln795_fu_495_p2 | xor | 0 | 0 | 2   | 1   | 2   |
| Total               |     | 0 | 0 | 294 | 266 | 173 |

## \* Multiplexer:

| Name                          | LUT | Input Size | Bits | Total Bits |
|-------------------------------|-----|------------|------|------------|
| ap_NS_fsm                     | 59  | 11         | 1    | 11         |
| i_fu_152                      | 9   | 2          | 31   | 62         |
| in_stream_TREADY_int_regslice | 9   | 2          | 1    | 2          |
| k_reg_208                     | 9   | 2          | 32   | 64         |
| out_stream_TDATA_blk_n        | 9   | 2          | 1    | 2          |
| x_localbuff_V_address0        | 14  | 3          | 6    | 18         |
| x_localbuff_V_ce0             | 14  | 3          | 1    | 3          |
| x_localbuff_V_we0             | 9   | 2          | 1    | 2          |
| Total                         | 132 | 27         | 74   | 164        |

## \* Register:

| Name                                                         | FF  | LUT | Bits | Const Bits |
|--------------------------------------------------------------|-----|-----|------|------------|
| add_ln27_reg_571                                             | 31  | 0   | 31   | 0          |
| add_ln37_reg_579                                             | 32  | 0   | 32   | 0          |
| ap_CS_fsm                                                    | 10  | 0   | 10   | 0          |
| col_size_read_reg_541                                        | 32  | 0   | 32   | 0          |
| grp_matVecMul_Pipeline_VITIS_LOOP_28_1_fu_219_ap_start_reg   | 1   | 0   | 1    | 0          |
| grp_matVecMul_Pipeline_matmul_inner_loop_fu_240_ap_start_reg | 1   | 0   | 1    | 0          |
| i_fu_152                                                     | 31  | 0   | 31   | 0          |
| k_reg_208                                                    | 32  | 0   | 32   | 0          |
| no_of_vectors_read_reg_552                                   | 32  | 0   | 32   | 0          |
| out_pkt_last_V_reg_589                                       | 1   | 0   | 1    | 0          |
| row_size_read_reg_547                                        | 32  | 0   | 32   | 0          |
| shl_ln_reg_594                                               | 6   | 0   | 13   | 7          |
| sub_reg_563                                                  | 32  | 0   | 32   | 0          |
| trunc_ln1169_reg_584                                         | 6   | 0   | 6    | 0          |
| Total                                                        | 279 | 0   | 286  | 7          |

## == Interface

## \* Summary:

| RTL Ports             | Dir | Bits | Protocol | Source Object | C Type |
|-----------------------|-----|------|----------|---------------|--------|
| s_axi_control_AWVALID | in  | 1    | s_axi    | control       | array  |
| s_axi_control_AWREADY | out | 1    | s_axi    | control       | array  |
| s_axi_control_AWADDR  | in  | 14   | s_axi    | control       | array  |
| s_axi_control_WVALID  | in  | 1    | s_axi    | control       | array  |
| s_axi_control_WREADY  | out | 1    | s_axi    | control       | array  |
| s_axi_control_WDATA   | in  | 32   | s_axi    | control       | array  |
| s_axi_control_WSTRB   | in  | 4    | s_axi    | control       | array  |
| s_axi_control_ARVALID | in  | 1    | s_axi    | control       | array  |
| s_axi_control_ARREADY | out | 1    | s_axi    | control       | array  |
| s_axi_control_ARADDR  | in  | 14   | s_axi    | control       | array  |
| s_axi_control_RVALID  | out | 1    | s_axi    | control       | array  |
| s_axi_control_RREADY  | in  | 1    | s_axi    | control       | array  |
| s_axi_control_RDATA   | out | 32   | s_axi    | control       | array  |
| s_axi_control_RRESP   | out | 2    | s_axi    | control       | array  |

|                      |     |    |            |                     |              |
|----------------------|-----|----|------------|---------------------|--------------|
| s_axi_control_BVALID | out | 1  | s_axi      | control             | array        |
| s_axi_control_BREADY | in  | 1  | s_axi      | control             | array        |
| s_axi_control_BRESP  | out | 2  | s_axi      | control             | array        |
| ap_clk               | in  | 1  | ap_ctrl_hs | matVecMul           | return value |
| ap_rst_n             | in  | 1  | ap_ctrl_hs | matVecMul           | return value |
| interrupt            | out | 1  | ap_ctrl_hs | matVecMul           | return value |
| in_stream_TDATA      | in  | 32 | axis       | in_stream_V_data_V  | pointer      |
| in_stream_TVALID     | in  | 1  | axis       | in_stream_V_dest_V  | pointer      |
| in_stream_TREADY     | out | 1  | axis       | in_stream_V_dest_V  | pointer      |
| in_stream_TDEST      | in  | 1  | axis       | in_stream_V_dest_V  | pointer      |
| in_stream_TKEEP      | in  | 4  | axis       | in_stream_V_keep_V  | pointer      |
| in_stream_TSTRB      | in  | 4  | axis       | in_stream_V_strb_V  | pointer      |
| in_stream_TUSER      | in  | 1  | axis       | in_stream_V_user_V  | pointer      |
| in_stream_TLAST      | in  | 1  | axis       | in_stream_V_last_V  | pointer      |
| in_stream_TID        | in  | 1  | axis       | in_stream_V_id_V    | pointer      |
| out_stream_TDATA     | out | 32 | axis       | out_stream_V_data_V | pointer      |
| out_stream_TVALID    | out | 1  | axis       | out_stream_V_dest_V | pointer      |
| out_stream_TREADY    | in  | 1  | axis       | out_stream_V_dest_V | pointer      |
| out_stream_TDEST     | out | 1  | axis       | out_stream_V_dest_V | pointer      |
| out_stream_TKEEP     | out | 4  | axis       | out_stream_V_keep_V | pointer      |
| out_stream_TSTRB     | out | 4  | axis       | out_stream_V_strb_V | pointer      |
| out_stream_TUSER     | out | 1  | axis       | out_stream_V_user_V | pointer      |
| out_stream_TLAST     | out | 1  | axis       | out_stream_V_last_V | pointer      |
| out_stream_TID       | out | 1  | axis       | out_stream_V_id_V   | pointer      |